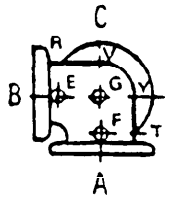
MIL-F-18180/9A
30 May 1986
SUPERSEDING
MIL-F-18180/9
30 April 1981

MILITARY SPECIFICATION SHEET

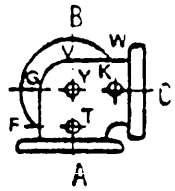
FLANGES AND FLANGED FITTINGS, PIPE, STEEL, CLASSES 150, 300, 400, 600, 900, 1500, AND 2500; 90 DEGREE, STRAIGHT AND REDUCING SIZE, SIDE OUTLET ELBOWS

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the 90 degree, straight and reducing size, side outlet elbows described herein shall consist of this document and the latest issue of specification MIL-F-18180.



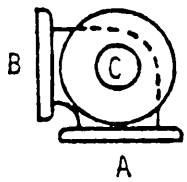
REAR VIEW
(RIGHT HAND)



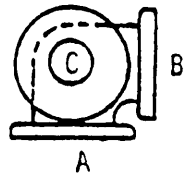
SIDE VIEW
(RIGHT HAND)



90° ELBOW



LEFT HAND



RIGHT HAND

FRONT VIEWS

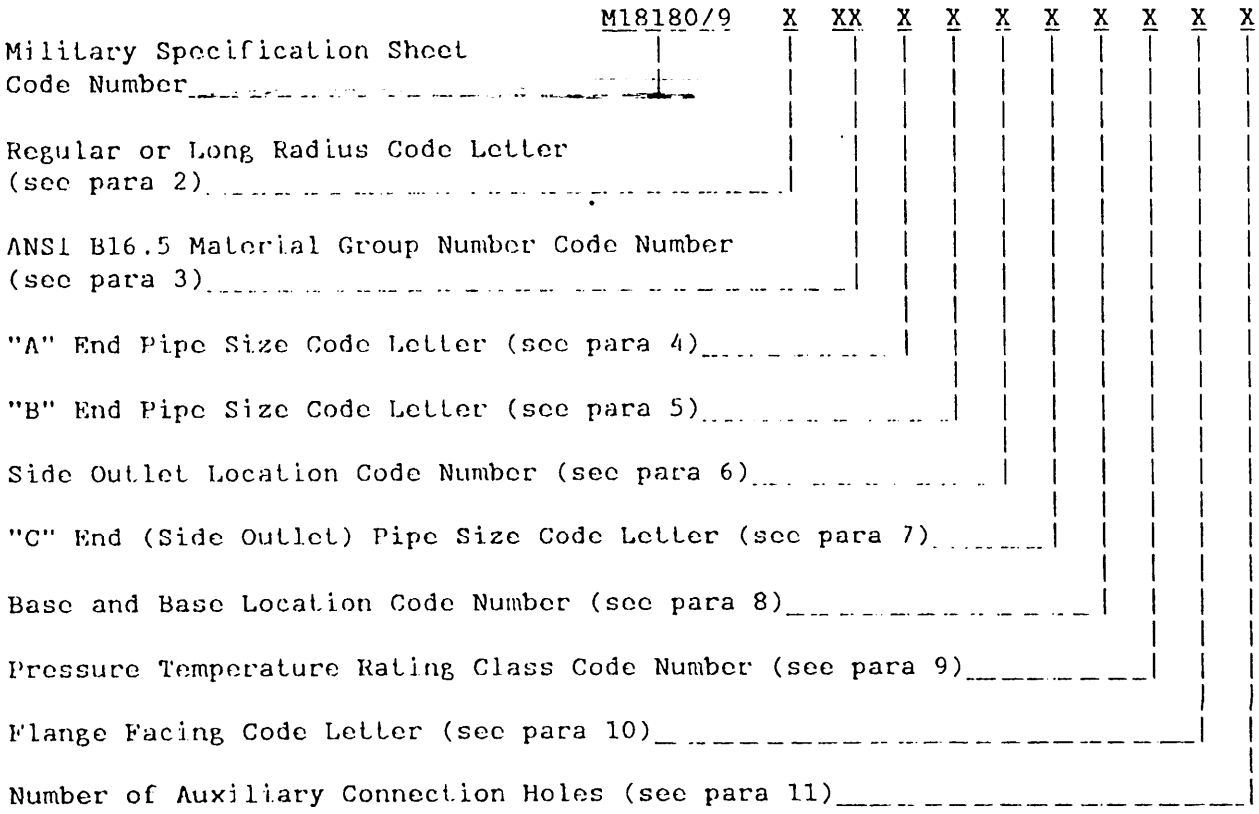
FIGURE 1. 90 degree side outlet elbows, straight and reducing size.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

AMSC N/A

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:04Z
Check the source to verify that this is the current version before use.

1. Military specification sheet part number. 90 degree, straight and reducing size, side outlet elbows covered by this specification sheet shall be designated in the following form (see 6.2(b) of MIL-F-18180).



2. Regular or long radius elbow. The radius of the elbow is designated by a single letter (see table 1).

TABLE I. Code letter to elbow radius.

Code Letter	Elbow Radius
R	Regular radius
L	Long radius 1/

1/ Long radius elbow only available in classes 150 and 300.

3. ANSI B16.5 material group number. The ANSI B16.5 Material Group No. for straight and reducing size, side outlet elbows is designated by two digits (see table II).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:04Z
Check the source to verify that this is the current version before use.

TABLE II. Code number to material group number.

Code Number	ANSI B16.5 Material Group No.	ANSI B16.5 Acceptable Material (ASTM Standards for)
01	Group 1.1	A105, A216 Grade WCB, A350 Grade LF2, A515 Grade 70, A516 Grade 70 or A537 Class 1.
02	Group 1.2	A203 Class B or E, A216 Grade WCC, A350 Grade LF3 or A352 Grade LC2 or LC3.
04	Group 1.4	A350 Grade LF1, A515 Grade 60 or A516 Grade 60.
05	Group 1.5	A182 Grade F1, A204 Grade A or B, A217 Grade WC1 or A352 Grade LC1.
07	Group 1.7	A182 Grade F2, A204 Grade C or A217 Grade WC 4 or WC 5.
09	Group 1.9	A182 Grade F11 or F12, A217 Grade WC6 or A387 Grade 11.
10	Group 1.10	A182 Grade F22, A217 Grade WC9 or A387 Grade 22.
13	Group 1.13	A182 Grade F5a or A217 Grade C5.
14	Group 1.14	A182 Grade F9 or A217 Grade C12.
21	Group 2.1	A182 Grade F304 or F304H, A240 Type 304 or A351 Grade CF3 or CF8.
22	Group 2.2	A182 Grade F316 or F316H, A240 Type 316 or 317 or A351 Grade CF3M or CF8M.
23	Group 2.3	A182 Grade F304L or F316L or A240 Type 304L or 316L.
24	Group 2.4	A182 Grade F321 or F321H or A240 Type 321 or 321H.
25	Group 2.5	A182 Grade F347, F347H, F348 or F348H, A240 Type 347, 347H, 348 or 348H or A351 Grade CF8C.
26	Group 2.6	A240 Type 309S or A351 Grade CH8 or CH20.
27	Group 2.7	A182 Grade F310, A240 Type 310S or A351 Grade CK20.

4. End "A" nominal pipe size The nominal pipe size for flanged end "A" (largest end) of straight and reducing size, side outlet elbows is designated by a single letter (see table III).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:04Z
Check the source to verify that this is the current version before use.

5. End "B" nominal pipe size. The nominal pipe size for flanged end "B" straight and reducing size, side outlet elbows is designated by a single letter (see table III).

TABLE III. Code letter to nominal pipe size.

Code Letter	Nominal pipe size (inches)
A	1/2
B	3/4
C	1
D	1-1/4
E	1-1/2
F	2
G	2-1/2
H	3
J	3-1/2
K	4
M	5
N	6
P	8
R	10
S	12
T	14
U	16
W	18
X	20
Y	24

6. Side outlet location. The location of the side outlet on straight and reducing size, side outlet elbows is designated by a single digit (see table IV).

TABLE IV. Code number to side outlet location.

Code Number	Side outlet location
1	Right hand
2	Left hand

7. End "C" (side outlet) nominal pipe size. The nominal pipe size for the flanged side outlet ("C" end) of straight and reducing size, side outlet elbows is designated by a single letter (see table III).

8. Base and base location. Whether straight or reducing size, side outlet elbows have a support base and, if so, where the base is located is designated by a single digit (see table V).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:04Z
Check the source to verify that this is the current version before use.

TABLE V. Code number to base and base location

Code Number	Base and Base Location 1/
7	Without base
8	With base (for straight sizes)
1	With base opposite small end
2	With base opposite large end

1/ Base elbows are not available in class 2500 and in 1/2 inch through 1-1/2 inch nominal pipe size of classes 150 through 1500.

9. Pressure-temperature rating. The pressure temperature rating class of flanged, straight and reducing size, side outlet elbows is designated by a single digit (see table VI).

TABLE VI. Code number to class.

Code Number	Class
1	150
3	300
4	400 1/
6	600
9	900
5	1500
2	2500

1/ Use only for nominal pipe size 4 inch through 24 inch; 1/2 inch through 3-1/2 inch nominal pipe size shall be rated as class 600 since in these sizes all dimensions for class 400 and class 600 are exactly the same.

10. Flange facing. The facing of the flanged ends is designated by a single letter (see table VII).

TABLE VII. Code letter to flange facing.

Code Letter	Flange Facing
A	Raised
B	Flat
C	Ring-Joint
D	Large Male
E	Large Female
F	Small Male
G	Small Female
H	Large Tongue
J	Small Tongue
K	Large Groove
M	Small Groove

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:04Z
Check the source to verify that this is the current version before use.

11. Auxiliary connections. The number of auxiliary connection holes in the flanged fittings is identified by a single digit.

11.1 No connections. If there are no auxiliary connection holes, this digit is zero (0).

11.2 Number of connections. If there are auxiliary connection holes, this digit is the numerical value of the number of auxiliary connection holes (1 through 9).

11.2.1 Type and location of connections. If there are auxiliary connection holes, the holes are listed after the number of auxiliary connection holes digit and each is identified by a two-letter code indicating type of connection (B for butt-weld, S for socket weld and T for threaded) and location of connection (E, F, G, K, R, S, T, V, W or Y or E, K, M, N, R, S, T, V, W or Y for elbows without base; E, F, G, K, R, T or W or E, K, M, N, R, T or W for elbows with base).

11.2.2 Size of connections. If the auxiliary connection holes are not the standard size specified in ANSI B16.5, the size is prefixed to the two-letter type and location code (i.e., 3 TE, 7/8 SH).

12. Military specification part number definition. The military specification sheet part number is a definitive part number which is formed by combining the military specification sheet code number (M18180/9 for MIL-F-18180/9) the regular or long radius code letter, the ANSI B16.5 material group number code number, the "A" end pipe size code letter, the "B" end pipe size code letter, the side outlet location code number, the "C" end (side outlet) pipe size code letter, the base and base location code number, the pressure temperature rating class code number, the flange facing code letter and the number of auxiliary connection holes as shown in paragraph 1.

13. Military specification part number limitations. The military specification sheet part number system requires that all flanged ends on the fitting have the same flange facing, has only one (1) side outlet and that there are not more than nine (9) auxiliary connections.

Custodians:

Army - ME
Navy - YD

Preparing Activity:

Navy - YD

(Project 4730-0712)

Review Activity:

DLA - CS

User Activities:

Navy - CG, MC

Source: https://assist.dla.mil -- Downloaded: 202640-18T11:04Z
Check the source to verify that this is the current version before use.